

Protect Your System Against Shellshock

This informative article explains how the newly discovered Shellshock bug operates. It gives simple, easy-to-implement methods to reduce your system's vulnerability to it as well as minimise the harmful effects.



Shellshock is a security bug, which allows the attacker to execute arbitrary commands in the UNIX bash shell, a command line interface commonly used in UNIX and Linux distributions. This bug was first discovered by Stéphane Chazelas on September 12, 2014, and he initially called it 'bashdoor'. It was assigned the CVE (common vulnerability and exposures) identifier 'CVE-2014-6271', and publicly disclosed on September 24, 2014. This article will discuss the effects of the bug, ways to test your system's vulnerability to it and also explore ways to protect yourself against attackers. Similar vulnerabilities are being discovered continuously and patches are being updated regularly.

Test your system

To test if your system is vulnerable to the Shellshock bug or not, open up a terminal (*Terminal.App* for Mac OS X) and type the following command:

```
env x='()' { ;; } echo vulnerable' bash -c 'echo hello'
env x='()' { ;; } echo vulnerable' sh -c 'echo hello'
```

Here, we are setting an environment variable 'x'. The part *echo*

vulnerable is the arbitrary command that is being executed before the actual bash command, i.e., *bash -c 'actual command'*, hence giving the attacker a chance to run arbitrary commands on your system.

You can further test your system for the vulnerability (as reported by Tavis Ormandy - CVE-2014-7169) by running the following command:

```
env X='()' { (a)=\ bash -c "echo date"; cat echo
env X='()' { (a)=\ sh -c "echo date"; cat echo
```

You will notice that the date is displayed and you can also find a file named 'echo' in the current directory. This means that your system is still vulnerable.

Here is another test that you can run, which will print out 'not patched' if your system is not yet patched or is still vulnerable to the bug:

```
env foo='()' { echo not patched; }' bash -c foo
```

The widespread effects of Shellshock

The Shellshock bug potentially affects all the systems that